# What is an ecosystem?

An ecosystem is a complex **system** that includes all the plants, animals and non-living things found in a certain place, as well as all the different ways those things **interact**. We say that something interacts with something else when it acts in a way that affects the other thing.

For example, a garden pond contains living things – also called **organisms** – such as frogs, snails, fish, water birds, water plants and algae, as well as non-living things such as fresh water, rocks, sand, air, sunlight and soil. The garden pond ecosystem is all of these living and non-living things as well as all of the ways these things interact and depend on each other – from a fish eating a snail to water lilies using sunlight to make food. All the living things in an ecosystem are known as a **community**. The variety of living things in the ecosystem is known as the ecosystem's **biodiversity**.

☞ **1** Draw a line between each word and its meaning.

| Word | Meaning |
|---|---|
| **a)** ecosystem | the variety of different lifeforms in a particular place or ecosystem |
| **b)** system | all the living things that live in the same place and depend on one another |
| **c)** interact | act in ways that affect one another |
| **d)** organism | the living and non-living things in a certain place and all their interactions |
| **e)** community | a set of connected things that form a whole |
| **f)** biodiversity | a living thing |

**2** Think of the ways that animals, plants and non-living things interact in a garden pond ecosystem. List four interactions in the answer spaces below, and then share your answers with a partner. One has been done for you.

A frog eats an insect.

_____ _____

_____ _____

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# What is an ecosystem? CONT'D

### *Balanced ecosystems*

An ecosystem is described as balanced, or **healthy**, when all the groups of living things are **stable** and nothing is disturbing the flow of energy and nutrients. Human activities – such as cutting down trees, introducing new animal species and planting crops that are not native to an area – can affect the delicate balance of an ecosystem. Other things that may affect ecosystems are floods, fires, drought and changes in climate.

**3** Fill in the table by giving examples of how these things might disturb the balance in an ecosystem. One has been done for you.

| **Disturbance** | **Possible effect** |
|---|---|
| cutting down trees | destroys the homes of some animals, causing them to move away or die out |
| introducing new plant or animal species | |
| flood | |
| drought | |


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